FLC score math and all tuning knobs

1) What changed (old → new)

Old score (what caused your repeats)

You were effectively using:

$$S_{old}(r) = \frac{1}{1 + D^2(r) + P(r)}$$

Problem: if a recipe becomes a near-perfect match, then $D^2(r) \to 0$. At that point, the penalty has to be *huge* to compete, and because your penalty mostly depended on **recency** (and decayed fast), the recipe always "comes back" on a regular cycle.

New score (what fixes the compensation problem)

You split the score into:

Base similarity (fit):

$$S_0(r) = \frac{1}{1 + D^2(r)}$$

Suppression (diversity control):

$$S_{new}(r) = S_0(r) \cdot \exp\big(-\gamma\, P(r)\big)$$

Key change: penalty is now **multiplicative** (a gate), not additive. That means a recipe can't "buy its way back" purely by having tiny distance.

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2) Distance term $D^2(r)$ (closeness to target)

$$D^2(r) = \sum_{i \in \{cal, prot, carb, fat, prep, spice\}} w_i \left(\frac{r_i - T_i}{\sigma_i}\right)^2$$

Variables

- r: a candidate recipe.
- i: one feature dimension.
- r_i: recipe r's value on feature i (e.g., protein grams).
- T_i: target value on feature i output by your fuzzy layer (after defuzzification).
- σ_i: scale (standard deviation) for feature i used to normalize magnitudes.
- w_i: weight for feature i. Higher w_i = that feature matters more.

What changes D^2

- Increasing w_i makes mismatch on feature i hurt more.
- Increasing σ_i makes mismatch on feature i hurt less (because you divide by σ_i).

Important: D^2 controls *taste/fit*, not diversity.

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3) Penalty term $P(r)$ (diversity control)

You need **recency + frequency**.

3.1 Recipe-level penalty

$$P_{recipe}(r) = A_r \, \exp\left(-\frac{d_r}{\tau_r}\right) \, (1 + c_{r,W})^{p_r}$$

Variables - d_r: days since recipe r was last consumed. - $c_{r,W}$: count of how many times recipe r occurred in the last W days. - A_r: recipe penalty amplitude. - τ_r: recipe penalty decay time constant (in days). - p_r: frequency exponent (how aggressively repeats stack). - W: frequency window length (days).

3.2 Cuisine/cluster penalty

$$P_{cuisine}(r) = A_c \, \exp\left(-\frac{d_c}{\tau_c}\right) \, (1 + c_{c,W})^{p_c}$$

Variables - d_c: days since the cuisine/cluster for recipe r was last consumed. - $c_{c,W}$: count of that cuisine/cluster in last W days. - A_c: cuisine amplitude. - τ_c: cuisine decay. - p_c: cuisine frequency exponent.

3.3 Total penalty

$$P(r) = P_{recipe}(r) + P_{cuisine}(r)$$

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4) Final score (what you rank on)

$$S(r) = \frac{1}{1 + D^2(r)} \cdot \exp\big(-\gamma \, P(r)\big)$$

Variables

- γ: suppression strength (master knob). Bigger γ = penalties hit harder.

Log form (recommended for numerical stability)

$$\log S(r) = -\log(1 + D^2(r)) - \gamma \, P(r)$$

Rank by largest $\log S$.

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5) Selection (to avoid deterministic cycles)

If you always take argmax, you will get repeating schedules.

Top-K + softmax sampling

1) compute $\log S(r)$ for all candidates 2) take Top-K recipes 3) sample:

$$\Pr(r) = \frac{\exp(\beta \log S(r))}{\sum_{j \in TopK} \exp(\beta \log S(j))}$$

Variables: - K: shortlist size. - β: determinism (temperature). High $\beta \approx$ greedy; low $\beta \approx$ more random.

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6) One hard constraint (optional but effective)

Hard block exact repeats on consecutive days: - if $d_r \leq 1$: set $S(r) = 0$

Variable: - $hard_block_days$ (0, 1, 2...)

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7) The knobs and what they do

Diversity knobs (the real “space repetition” dials)

- γ: master suppression strength.
- ↑ $\gamma \Rightarrow$ repeats pushed out longer.
- ↓ $\gamma \Rightarrow$ repeats come back sooner.
- τ_r, τ_c: decay speed.
- ↑ $\tau \Rightarrow$ penalty lasts longer (longer cooldown tail).
- ↓ $\tau \Rightarrow$ penalty disappears quickly (short cycles).
- W: frequency memory window.
- ↑ $W \Rightarrow$ “repeat count” stays relevant longer.
- ↓ $W \Rightarrow$ system forgets repeats sooner.
- p_r, p_c: stacking aggressiveness.
- ↑ $p \Rightarrow$ 2nd/3rd time in window gets punished much more.
- ↓ $p \Rightarrow$ stacking is weak (more repeats).
- A_r, A_c: base amplitude.

- ↑ $A \Rightarrow$ stronger immediate push away after consumption.
- ↓ $A \Rightarrow$ softer push.
- **hard_block_days**: hard constraint.
- 1 is usually enough.

Taste/fit knobs (how strictly you match the fuzzy target)

- w_i: feature importance.
- ↑ w on protein means you'll sacrifice variety to keep protein close.
- σ_i: scaling.
- If spice has wrong σ, spice mismatch becomes meaningless or dominates.

Randomness knobs (when you later add user swaps / exploration)

- β: determinism.
- ↑ $\beta \Rightarrow$ more greedy.
- ↓ $\beta \Rightarrow$ more exploration.
- K: candidate pool size.
- ↑ $K \Rightarrow$ more variety available to sampling.

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8) How to tune to a desired repeat interval (practical)

Pick the repeat interval you want (say 14–21 days for recipes).

1) Fix taste weights w_i first (don't tune diversity before taste is sane). 2) Set hard block to 1 day. 3) Increase W to about the interval you want to control (e.g., 14). 4) Increase τ_r until score recovery takes roughly weeks, not days. 5) Increase γ if repeats still return too soon. 6) Increase p_r if you see "twice in the last 14 days" still happening too often. 7) If you add user swaps later, you can relax γ and/or τ_r because the user can correct the system.